NAMANGA SECONDARY SCHOOL

FORM FOUR — INFORMATION AND COMPUTER STUDIES
DATABASE PRACTICAL EXAMINATION

Time: 3 Hours **Total Marks:** 100

GENERAL INSTRUCTIONS

1. Use Microsoft Access 2021 to complete the practical examination.
2. Create and save the database as **NAMANGA_STUDENT_DATABASE**.
3. Create exactly three tables: **STUDENT**, **CLASS**, and **ACTIVITY**.
4. Enter only the records supplied in this question paper.
5. Create only the relationship between CLASS and STUDENT. Do not create a relationship involving ACTIVITY.
6. Use Design View, Query Design, Lookup Wizard, Forms and Reports as instructed.
7. Save every database object using the exact names given.
8. Check the database carefully before submission.

A. CREATE THE STUDENT TABLE — 15 MARKS

Create a table named **STUDENT** with the following fields and appropriate data types. Set StudentID as the primary key.

Field Name	Data Type	Key / Requirement
StudentID	Short Text	Primary Key
FirstName	Short Text	
LastName	Short Text	
Gender	Short Text	Lookup: Male, Female
DOB	Date/Time	
ClassCode	Short Text	Lookup from CLASS.ClassCode
Phone	Short Text	
Mark	Number	Validation: Between 0 And 100

StudentID	FirstName	LastName	Gender	DOB	ClassCode	Phone	Mark
ST001	Amani	Mollel	Male	12/03/2010	F2A	0712345678	78
ST002	Neema	Joseph	Female	27/07/2011	F2A	0756789012	84
ST003	Musa	Kweka	Male	19/01/2009	F3A	0765432109	67
ST004	Halima	Said	Female	05/11/2010	F2B	0687654321	72
ST005	Baraka	Mushi	Male	22/09/2008	F3A	0711223344	91
ST006	Maria	John	Female	14/02/2012	F1A	0744556677	58
ST007	Kelvin	Mrema	Male	30/06/2011	F2B	0788990011	63
ST008	Rehema	Nnko	Female	17/08/2009	F3B	0622334455	88
ST009	Omari	Juma	Male	02/05/2010	F2A	0699887766	49
ST010	Asha	Mwakalinga	Female	25/12/2008	F3B	0714556677	76

B. CREATE THE CLASS TABLE — 10 MARKS

Create a table named **CLASS**. Set ClassCode as the primary key. All fields are Short Text.

ClassCode	ClassName	Stream	RoomNo	ClassTeacher
F1A	Form I A	A	R01	Mr Kato
F2A	Form II A	A	R02	Ms Neema
F2B	Form II B	B	R03	Mr Baraka
F3A	Form III A	A	R04	Ms Asha
F3B	Form III B	B	R05	Mr Musa

C. CREATE THE ACTIVITY TABLE — 10 MARKS

Create a table named **ACTIVITY**. Set ActivityID as the primary key.

ActivityID	ActivityName	Category	ActivityDate	Participants	Fee
AC01	Cyber Safety Talk	ICT	20/01/2026	80	0
AC02	Football Match	Sports	24/01/2026	96	500
AC03	Database Club	Academic	28/01/2026	24	0
AC04	Science Fair	Academic	13/02/2026	55	1000
AC05	Girls in ICT Day	ICT	28/02/2026	45	0
AC06	Netball Match	Sports	07/03/2026	60	300
AC07	ICT Workshop	ICT	21/03/2026	38	2000
AC08	Programming Challenge	ICT	28/03/2026	27	500

D. KEYS, LOOKUPS, VALIDATION AND RELATIONSHIP — 15 MARKS

• Set StudentID, ClassCode (in CLASS), and ActivityID as primary keys where specified.

• Gender must use a Lookup list containing only **Male** and **Female**; Limit To List = Yes.

• ClassCode in STUDENT must use a Lookup list from CLASS.ClassCode; Limit To List = Yes.

• Set STUDENT.Mark validation to **Between 0 And 100** with an appropriate validation message.

• Set STUDENT.Phone validation so it accepts exactly 10 digits beginning with 0.

• Set ACTIVITY.Participants validation to **>=1** and Fee validation to **>=0**.

• Set ActivityDate to accept dates in the year 2026.

• Create only one relationship: CLASS.ClassCode → STUDENT.ClassCode. Enforce Referential Integrity. Do not relate ACTIVITY.

E. DIFFICULT QUERIES — 20 MARKS

Important: Create **FOUR QUERIES ONLY**. Each query is deliberately challenging and must be saved using the exact name given.

QUERY I — Q_ICT_ACADEMIC_PERIOD

Using ACTIVITY, display ActivityID, ActivityName, Category, ActivityDate, Participants and Fee for activities held from 01/02/2026 to 31/03/2026 where the Category is either ICT or Academic. Apply the correct AND/OR criteria grouping so that the date condition applies to both categories. Sort ActivityDate in ascending order.

QUERY II — Q_HIGH_COLLECTION

Using ACTIVITY, create a calculated field named TotalCollection that calculates Participants × Fee. Display ActivityName, Category, Participants, Fee and TotalCollection. Show only records where TotalCollection is greater than 50,000, and sort TotalCollection from highest to lowest.

QUERY III — Q_FORMIII_PERFORMANCE

Create a query using both STUDENT and CLASS. Display FirstName, LastName, ClassName, Stream, Mark and a calculated field named PerformanceBand. PerformanceBand must return Excellent for marks ≥80, Good for marks from 60 to 79, and Needs Improvement for marks below 60. Show only Form III students and sort Mark from highest to lowest.

QUERY IV — Q_CLASS_GENDER_CROSSTAB

Create a Crosstab Query using STUDENT and CLASS. Show ClassName as row headings, Gender as column headings, and Count of StudentID as the value. The result must show how many male and female students are in each class.

F. ADVANCED SORTING AND FILTERING — 10 MARKS

1. Using STUDENT, create a query named **Q_TOP5_MARKS** that displays the five students with the highest marks. Sort Mark descending and use the Top Values property correctly.

2. Create a query named **Q_FORMII_LOW_OR_HIGH** that displays students in F2A or F2B whose marks are either below 60 or 80 and above. Use correct AND/OR grouping.

G. FORMS AND REPORTS — 15 MARKS

Create the following database objects:

Object	Requirement
FORM_STUDENT	Form based on STUDENT; Columnar layout.
FORM_CLASS	Form based on CLASS; Tabular layout.
FORM_ACTIVITY	Form based on ACTIVITY; Justified layout.
REPORT_STUDENT_CLASS	Based on Q_FORMIII_PERFORMANCE; group by ClassName; sort Mark descending.
REPORT_ACTIVITY_CATEGORY	Based on ACTIVITY; group by Category; display ActivityName, Participants and Fee.
REPORT_PERFORMANCE	Based on Q_FORMIII_PERFORMANCE; display student name, Mark and PerformanceBand; inclu

H. TESTING AND FINAL CHECK — 15 MARKS

• Test Mark = 105. The value must be rejected.

• Test Participants = 0. The value must be rejected.

• Test Fee = -500. The value must be rejected.

• Test Gender = Other. The value must be rejected.

• Test Phone = 07123456. The value must be rejected.

• Test ClassCode = F9A. The value must be rejected because of the lookup/relationship rules.

• After testing, restore valid data and confirm that all four difficult queries, forms, reports, keys, validation rules and the relationship work correctly.

• Save the final database as **NAMANGA_STUDENT_DATABASE** and submit it.

END OF EXAMINATION